Samuel Anes

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Bachelor of Science in Electrical and Computer Engineering — Graduating May 2030
Selected Coursework: Intro to Electrical and Computer Engineering, Calculus in Three Dimensions, Concepts of Math

Homeschool, Stanford Online High School (Graduated June 2026) — Sudbury, MA
GPA: Weighted: 4.4 / Unweighted: 3.97 — SAT: 1530 ACT: 35
Selected Coursework: AP Physics C: Electricity & Magnetism, AP Physics C: Mechanics, Linear Algebra, Harvard CS50x

SKILLS

Programming: Python, Java, C, C++, SQL, JavaScript, HTML, CSS, Svelte
Software & Tools: GitHub, Claude Code, ROS2, Verilog, Linux, Android Studio, OnShape
Hardware: Arduino, 3D Printers, CNC Machines, Multimeter, Oscilloscope

ENGINEERING EXPERIENCE

Carnegie Mellon Formula SAE Electric Team (September 2026-Present) — Pittsburgh, PA
Driverless Controls Subteam Member — September 2026-Present

- Developed a regenerative braking algorithm and validated it against real-world vehicle data using Python and Matplotlib, informing controls development for the FSAE Endurance event.

FIRST Tech Challenge Team #2845 (September 2022-April 2026) — Tampa, FL
Team Captain | FIRST Dean's List Semi-Finalist — September 2024-April 2026

- Rebuilt the team's Java software architecture into a modular command-based framework supporting sequential and parallel execution, enabling concurrent subsystem control during autonomous routines.
- Developed three-wheel + IMU odometry with periodic computer vision sensor fusion corrections and integrated the localization with a PID-based point-to-point robot navigation for autonomous field movement.
- Developed OpenCV computer-vision pipelines using HSV color segmentation, contour filtering, and geometric constraints to detect game elements and estimate their position relative to the robot.
- Developed a data-driven launcher velocity model by regressing target distance against measured AprilTag image area, automatically selecting launch velocity for autonomous scoring across varying distances.

Sabacc: Open Source Multiplayer Full-Stack Web Application
Project Founder and Leader — April 2024-Present

- Built and deployed a real-time multiplayer web application serving 1,000+ users, using Svelte, Flask, SQL, and WebSockets for reactive UI, backend game logic, persistence, and low-latency gameplay.
- Architected an object-oriented backend supporting multiple game variants through shared abstractions, and migrated between SQLite and PostgreSQL while maintaining persistent game data.

Embedded Projects

- Built an Arduino-based remote-control sailboat using XBee wireless communication, servo actuation, custom perfboard electronics, and 3D-printed mechanical components.
- Designed a capacitive-touch Arduino speedcubing timer with sensor calibration and 3D-printed enclosure.

EXTRACURRICULARS

Eagle Scout, Scouting America — January 2019-Present
Amateur Extra, FCC Licensed Amateur Radio Operator — May 2022-Present
Competitor, World Cube Association — January 2023-Present
Black Belt, Aidan Murray Martial Arts — 2019-Present